

Interactive Soundscape Project

Chance and Sound

Title: A Chance Driven Sound Performance

Tool: Freesound Explorer

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Project Overview

For this project, I designed an interactive sound performance that uses chance operations to determine how sounds are triggered and layered. The system is inspired by the idea that music and sound can be structured not only through deliberate control but also through randomness.

The performance uses a simple rule-based method involving a six-sided dice. Each roll of the dice determines the next action that the performer must execute within the sound interface. These actions may involve selecting a sound, layering sounds together, stopping a sound, or moving to a different area of the interface.

By introducing randomness into the decision-making process, the performer does not fully control the outcome of the piece. Instead, the final soundscape emerges from the interaction between the performer, the rule system, and the sound library.

The performance gradually evolves as more sounds are introduced and layered according to the dice results. Because the dice determines the actions, every performance produces a slightly different sonic outcome.

Software Used

I chose Freesound Explorer because it provides an interactive visual map of sounds from the Freesound database.

The interface groups similar sounds into clusters, making it easy to navigate through different sonic environments by moving the cursor across the screen. This spatial layout makes the platform ideal for an interactive performance where the performer needs to quickly select sounds from different categories.

Using this interface allowed me to focus on the design of the rule system while still exploring a wide range of environmental sounds. The visual clustering of sounds also helped structure the performance because the performer could move between different sound areas depending on the result of the dice.

Part 1: The Performance Rules

The performance is controlled by rolling a six-sided dice. Each number corresponds to a different action that must be performed within the sound interface.

The performer rolls the dice before each action and follows the rule associated with the number rolled.

RULES

Ending the Performance (One-Minute Rule)

Action:

The performer rolls the dice and follows the odd and even rules for a duration of one minute

The Rule:

During this one minute, the performer can roll the dice as many times as possible.

- If the number rolled is odd (1, 3, or 5), Sound A (Rain) is triggered.
- If the number rolled is even (2, 4, or 6), Sound B (Wind) is triggered.

Ending:

At the end of the one-minute performance, both Sound A (Rain) and Sound B (Wind) are played together, allowing the sounds to blend and form a final layered soundscape before the recording ends.

Why:

The one-minute rule keeps the performance structured while still allowing enough time for variation through the dice rolls. Playing both sounds together at the end creates a clear and intentional closing moment for the performance.

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Part 2: Why I Designed it This Way (Motivation)

My Inspiration

The main inspiration behind this project was the idea of using randomness as a creative tool. Instead of controlling every aspect of the performance, the system allows chance to influence the structure of the soundscape.

By using a dice as the decision-making mechanism, the performer becomes part of a system where both human interaction and randomness shape the final outcome.

The result is a performance that is unpredictable yet still structured by a clear set of rules.

Why I Chose This Software

I selected Freesound Explorer because its visual sound map allows quick and easy exploration of audio samples from the Freesound database.

The interface makes it easy to move between sound clusters and trigger samples rapidly, which is important for a performance that relies on responding to dice rolls. The visual layout also makes the experience feel more interactive and dynamic compared to traditional sound libraries.

Working With Limitations

One challenge when working with Free Sound Explorer is that sounds differ significantly in length and volume. Some samples are very short while others last much longer.

Design Principles

The project applies several interaction design principles such as constraints, feedback, and randomness.

The dice rules act as constraints that guide the performer's actions and prevent completely random interaction with the interface. Feedback is provided through the sounds themselves, as each action immediately changes the sound environment.

The use of randomness introduces variability into the system, ensuring that every performance produces a unique sonic result.

Reflection

This project demonstrates how simple rule systems can transform a sound browsing interface into an interactive musical instrument. By combining a chance-based decision system with environmental sound samples, the performance creates evolving soundscapes that differ each time they are performed.

The experiment highlights how randomness can be used as a creative design tool and how interaction design principles can influence sound-based digital performances.